

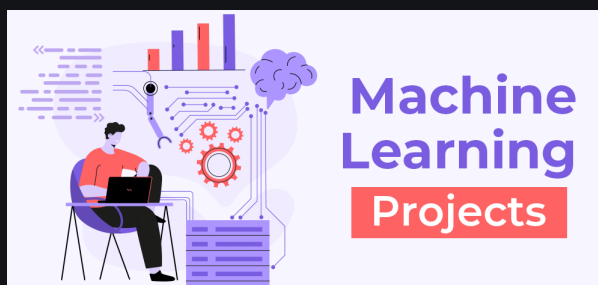
100+ Machine Learning Projects with Source Code [2024]

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Machine Learning gained a lot of popularity and become a necessary tool for research purposes as well as for Business. It is a revolutionary field that helps us to make better decisions and automate tasks. Machine Learning is the field of study that gives computers the capability to learn without being explicitly programmed.

In this article, you'll find the top **100+ latest Machine Learning projects and Ideas** which are beneficial for both beginners and as well experienced professionals. Whether you're a **final-year student** aiming for a standout resume or someone building a career, these machine learning projects provide hands-on experience, launching you into the exciting world of [Machine Learning](#) and [Data Science](#).



*Top Machine Learning Project with
Source Code [2024]*

We mainly include projects that solve real-world problems to demonstrate how machine learning solves these real-world

problems like: – Online Payment Fraud Detection using Machine Learning in Python, Rainfall Prediction using Machine Learning in Python, and Facemask Detection using TensorFlow in Python.

- [Machine Learning Project for Beginners](#)
- [Advanced Machine Learning Projects](#)

These projects provide a great opportunity for developers to apply their knowledge of machine learning and make an application that benefits society. By implementing these projects in data science, you be familiar with a practical working environment where you follow instructions in real time.

Machine Learning Project for Beginners in 2024 [Source Code]

Let's look at some of the best new machine-learning projects for beginners in this section and each project deals with a different set of issues, including supervised and unsupervised learning, classification, regression, and clustering. Beginners will be better prepared to tackle more challenging tasks by the time they have finished reading this article and have a better understanding of the fundamentals of machine learning.

Healthcare

Disease Prediction Using Machine Learning	Video
ML Heart Disease Prediction Using Logistic Regression	Video
Prediction of Wine type using Deep Learning	Video
Parkinson's Disease Prediction using Machine Learning in	Video

Python	
ML Kaggle Breast Cancer Wisconsin Diagnosis using Logistic Regression	Video
ML Cancer cell classification using Scikit-learn	Video
ML Kaggle Breast Cancer Wisconsin Diagnosis using KNN and Cross-Validation	Video
Autism Prediction using Machine Learning	Video

2. Finance and Economics

Credit Card Fraud Detection	Video
Dogecoin Price Prediction with Machine Learning	Video
Zillow Home Value (Zestimate) Prediction in ML	Video
Bitcoin Price Prediction using Machine Learning in Python	Video
Online Payment Fraud Detection using Machine Learning in Python	Video
Stock Price Prediction using Machine Learning in Python	Video
Stock Price Prediction Project using TensorFlow	Video
Microsoft Stock Price Prediction with Machine Learning	Video
Predicting Stock Price Direction using Support Vector Machines	Video

Share Price Forecasting Using Facebook Prophet	Video
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3. Food and Beverage

Wine Quality Prediction	Video
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4. Retail and Commerce

Sales Forecast Prediction – Python	Video
Customer Segmentation using Unsupervised Machine Learning in Python	Video
Analyzing selling price of used cars using Python	Video
Box Office Revenue Prediction Using Linear Regression in ML	Video
Flipkart Reviews Sentiment Analysis using Python	Video
Loan Approval Prediction using Machine Learning	Video
Loan Eligibility prediction using Machine Learning Models in Python	Video
House Price Prediction using Machine Learning in Python	Video
ML Boston Housing Kaggle Challenge with Linear Regression	Video

5. Sports

IPL Score Prediction Using Deep Learning	Video
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6. Health and Fitness

Calories Burnt Prediction using Machine Learning	V i d e o
Medical Insurance Price Prediction using Machine Learning in Python	V i d e o

7. Transportation and Traffic

Vehicle Count Prediction From Sensor Data	V i d e o
Ola Bike Ride Request Prediction	V i d e o

Skip to content

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8. Environmental Science

Rainfall Prediction using Machine Learning in Python	V i d e o
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9. Text and Image Processing

Detecting Spam Emails Using Tensorflow in Python	V i d e o
SMS Spam Detection using TensorFlow in Python	V i d e o

Share Your Experiences

simple machine learning model

Machine Learning Model Evaluation

Steps to Build a Machine Learning Model

Machine Learning Deployment

Machine learning deployment

Deploy your Machine Learning web app (Streamlit) on Heroku		o
	Python Classify Handwritten Digits with Tensorflow	V i d e o
	OCR of Handwritten digits OpenCV	V i d e o
	Recognizing HandWritten Digits in Scikit Learn	V i d e o
	Identifying handwritten digits using Logistic Regression in PyTorch	V i d e o
	Cartooning an Image using OpenCV – Python	V i d e o
	Count number of Object using Python-OpenCV	V i d e o
	Count number of Faces using Python – OpenCV	V i d e o
	Text Detection and Extraction using OpenCV and OCR	V i d e

	<u>o</u>
<u>CIFAR-10 Image Classification in TensorFlow</u>	<u>V</u> <u>i</u> <u>d</u> <u>e</u> <u>o</u>
<u>Black and white image colorization with OpenCV and Deep Learning</u>	<u>V</u> <u>i</u> <u>d</u> <u>e</u> <u>o</u>
<u>Handwritten Digit Recognition using Neural Network</u>	<u>V</u> <u>i</u> <u>d</u> <u>e</u> <u>o</u>

10. Social Media and Sentiment Analysis

<u>Twitter Sentiment Analysis using Python</u>	<u>V</u> <u>i</u> <u>d</u> <u>e</u> <u>o</u>
<u>Facebook Sentiment Analysis using python</u>	<u>V</u> <u>i</u> <u>d</u> <u>e</u> <u>o</u>

11. Other Important Machine Learning Projects

<u>Human Scream Detection and Analysis for Controlling Crime Rate</u>	<u>V</u> <u>i</u> <u>d</u> <u>e</u> <u>o</u>
<u>Spaceship Titanic Project using Machine Learning in Python</u>	<u>V</u> <u>i</u> <u>d</u> <u>e</u> <u>o</u>

Inventory Demand Forecasting using Machine Learning in Python	Video
Waiter's Tip Prediction using Machine Learning	Video
Fake News Detection using Machine Learning	Video
Fake News Detection Model using TensorFlow in Python	Video
Predict Fuel Efficiency Using Tensorflow in Python	Video

Advanced Machine Learning Projects With Source Code [2024]

We have discussed a variety of complex machine-learning ideas in this section that are intended to be challenging for users and span a wide range of topics. These subjects involve creating deep learning models, dealing with unstructured data, and instructing sophisticated models like convolutional neural networks, gated recurrent units, large language models, and reinforcement learning models.

1. Image and Video Processing

Multiclass image classification using Transfer learning	Video
Image Caption Generator using Deep Learning on Flickr8K dataset	Video
FaceMask Detection using TensorFlow in Python	Video
Dog Breed Classification using Transfer Learning	Video
Flower Recognition Using Convolutional Neural Network	Video
Cat & Dog Classification using Convolutional Neural Network in Python	Video
Traffic Signs Recognition using CNN and Keras in Python	Video
Residual Networks (ResNet) – Deep Learning	Video
Lung Cancer Detection using Convolutional Neural Network (CNN).	Video
Lung Cancer Detection Using Transfer Learning	Video
Black and white image colorization with OpenCV and Deep Learning	Video
Pneumonia Detection using Deep Learning	Video
Detecting Covid-19 with Chest X-ray	Video
Detecting COVID-19 From	

Chest X-Ray Images using CNN	Video
Image Segmentation Using TensorFlow	Video

2. Recommendation Systems

Ted Talks Recommendation System with Machine Learning	Video
Python Implementation of Movie Recommender System	Video
Movie recommendation based on emotion in Python	Video
Music Recommendation System Using Machine Learning	Video

3. Speech and Language Processing

Speech Recognition in Python using Google Speech API	Video
Voice Assistant using python	Video
Next Sentence Prediction using BERT	Video
Hate Speech Detection using Deep Learning	Video
Fine-tuning the BERT model for Sentiment Analysis	Video
Sentiment Classification Using BERT	Video
Sentiment Analysis with Recurrent Neural Networks (RNN).	Video

Autocorrect Feature Using NLP In Python	Video
Python NLP analysis of Restaurant reviews	Video
Restaurant Review Analysis Using NLP and SQLite	Video

4. Health and Medical Applications

Skin Cancer Detection using TensorFlow	Video
Heart Disease Prediction using ANN	Video

5. Security and Surveillance

Intrusion Detection System Using Machine Learning Algorithms	Video
License Plate Recognition with OpenCV and Tesseract OCR	Video
Detect and Recognize Car License Plate from a video in real-time	Video

6. Gaming and Entertainment

AI-Driven Snake Game using Deep Q Learning	Video
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7. Other Advanced Machine Learning Projects

Age Detection using Deep Learning in OpenCV	Video
Face and Hand Landmarks	Video

Detection using Python	
Human Activity Recognition – Using Deep Learning Model	Video
How can Tensorflow be used with the abalone dataset to build a sequential model?	Video

Machine Learning Projects – FAQs

What are some good machine-learning projects?

For beginners, recommended machine learning projects include sentiment analysis, sales forecast prediction, and image recognition.

How do I start an ML project?

To start a machine learning project, the first steps involve collecting data, preprocessing it, constructing data models, and then training those models with that data.

Which Language is used for machine learning?

Python and R are most popular and widely-used programming languages for machine learning.

Why do we need to build machine learning projects?

We need to build machine learning projects to solve complex problems,

automate tasks and improve decision-making.

What is the future of machine learning?

Machine learning is a fast-growing field of study and research, which means that the demand for machine learning professionals is also growing.

Are you passionate about data and looking to make one giant leap into your career? Our [Data Science Course](#) will help you change your game and, most importantly, allow students, professionals, and working adults to tide over into the data science immersion. Master state-of-the-art methodologies, powerful tools, and industry best practices, hands-on projects, and real-world applications. Become the executive head of industries related to **Data Analysis, Machine Learning**, and **Data Visualization** with these growing skills. Ready to Transform Your Future? ***Enroll Now to Be a Data Science Expert!***

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Machine Learning is an area of research that allows machines the ability to learn

🕒 3 min read

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Data Science For Beginners
Machine Learning
ML Maths
Data Visualisation
Pandas
NumPy
NLP
Deep Learning

	Python Tutorial	Computer Science	DevOps	System Design	School Subjects
	Python Programming Examples	GATE CS Notes	AWS	High Level Design	Mathematics
	Django Tutorial	Operating Systems	Docker	Design	Physics
	Python Projects	Computer Network	Kubernetes	Low Level Design	Chemistry
	Python Tkinter	Database Management	Azure	Design	Biology
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		Colleges			
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	DSA - Self Paced Course	JavaScript Full Course	Learning/Complete Machine Learning &	C Programming with Data Structures	DevOps Engineering
	DSA in JavaScript - Self Paced	React JS Course			AWS Solutions Architect

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DSA in Python	Django Web	Program -	Programming	Salesforc
- Self Paced	Development	[LIVE]	Course	Certified
C	Course	Data Analytics	Java	Administ
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Course Online	Bootstrap	Excel, SQL,	Course	
- Learn C with	Course	Python &	Python Full	
Data	Full Stack	PowerBI -	Course	
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Complete	- [LIVE]	Data Science		
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Preparation	Development	Program -		
Master	- [LIVE]	[LIVE]		
Competitive	Complete	Mastering		
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Core CS	Testing	and ChatGPT		
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Interview	Android	Course with		
Preparation	Mastery with	IBM		
Mastering	Kotlin [LIVE]	Certification		
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Design: LLD to				
HLD				
Tech				
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Preparation				
Crash Course				
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